



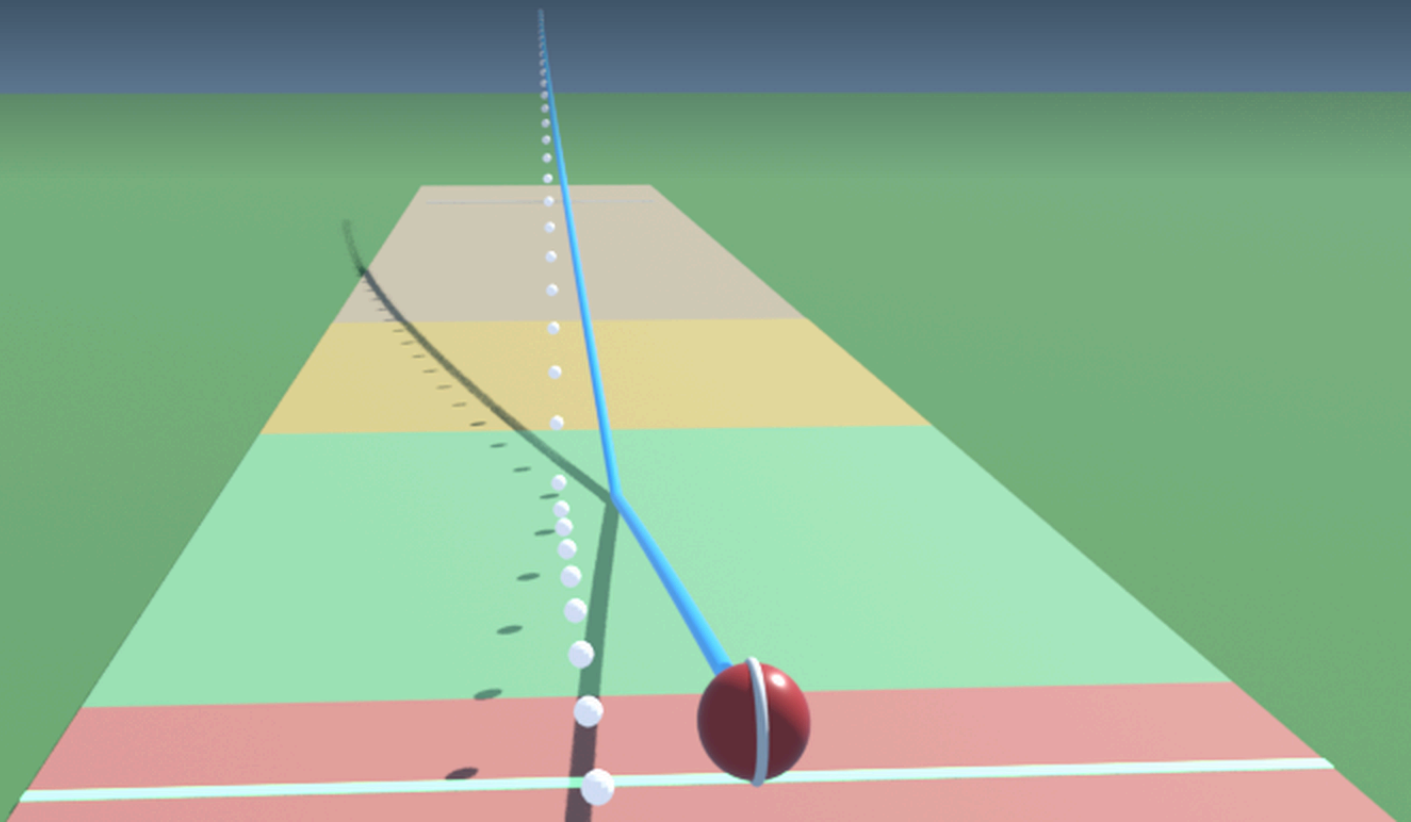
Thinking in Systems
NOT CODE

FIELD NOTE · CRICKET

Bowled by a Late Swinger

I got out to a ball that pitched on middle and hit off stump. So I modelled it — and found the movement was never the part I should have been trying to predict.

Ankit Naik · Thinking in Systems · August 2026



If you read nothing else

- **The finding** — length barely changes how much an outswinger moves. Full, good or short, it swings the same **0.38 m**. What length changes is how long you get to read it.
- **Why it matters** — you are not judging a variable amount of movement. You are judging a starting line and adding a near-constant offset.
- **The mechanism** — the sideways force is aerodynamic. It is set by pace and seam angle. It does not know where the ball bounced.
- **Do this** — meet it further out. Take guard outside the crease, stride to the pitch of the ball, or both — it halves the movement that arrives after you have committed.

0.38 m

Sideways movement — the same at every length

96 mm

The ball that bowled me: middle to off stump, after pitching

2×

Late movement you take on by playing back instead of at the pitch

SO WHAT

Stop trying to predict *how much* it will swing. Predict the line it started on, add about a ball and a half towards the off side, and meet it as far out as you can so that less of the movement happens after you have committed.

01 Length changes the timing, not the total

My assumption walking off: a fuller ball pitches closer to me, so there is less room left for it to move. Pitch it up and the problem shrinks.

LENGTH	PITCHES FROM STUMPS	TOTAL SWING	MOVES AFTER PITCHING	TIME TO READ IT
Overpitched	2.0 m	+0.39 m	+74 mm	81 ms
Good length	4.8 m	+0.38 m	+165 mm	191 ms
Short	8.0 m	+0.38 m	+261 mm	324 ms

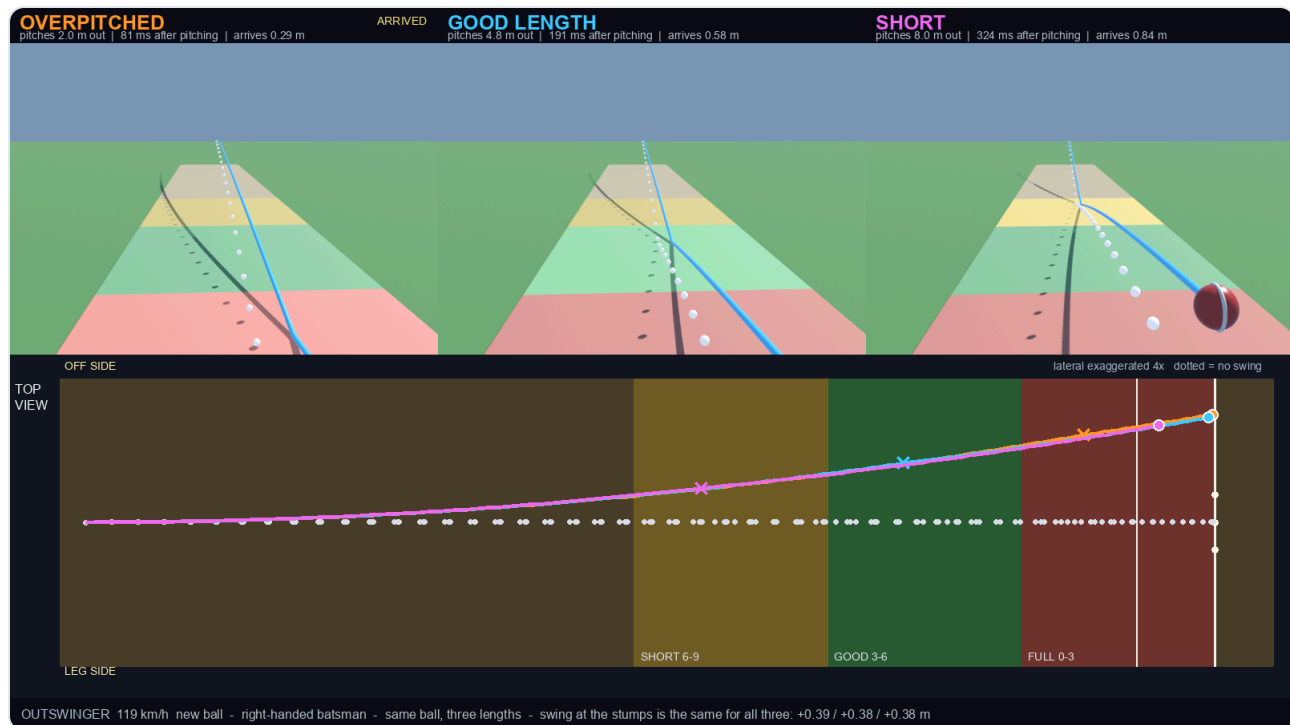


Figure 1. Notice the plan view at the bottom: the three coloured paths lie almost on top of each other. Only the crosses — where each ball pitches — separate. Half right: the fuller ball really does move less after pitching (74 mm against 261 mm), but it gives you 81 ms to see it instead of 324 ms. The *total* is the same either way.

02 Read the line, not the movement

If the movement is fixed, the only thing left to read is where the ball started. So I changed the release angle and left everything else alone.

RELEASED	PITCHES	REACHES THE STUMPS	SWING
-3.0°	0.45 m outside leg	0.49 m down the leg side	+0.38 m
-1.0°	on middle	off stump	+0.38 m
+1.5°	0.55 m outside off	0.82 m outside off, to slip	+0.38 m

- The release angle slides the whole line sideways. It does not change the swing.
- The middle row is the ball that bowled me — aimed one degree inside the stumps.
- So the movement is an offset you can learn, not a variable you must guess.

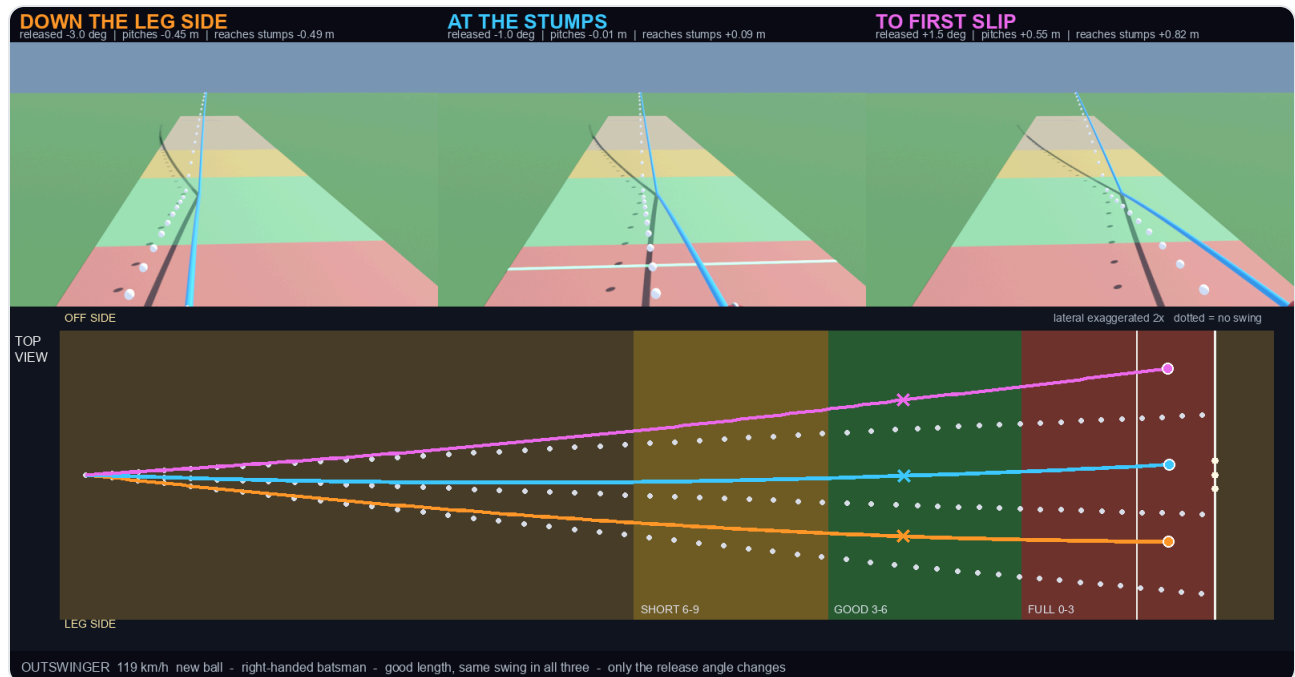


Figure 2. Three aims, three parallel curves. Each solid path sits the same distance from its own dotted twin — the twin being the same ball with the swing switched off.

03 Where to meet it

Movement accumulates with distance flown. So the question is not how much it swings — it is how much is still to come once you have committed.

MEET THE BALL	AT	MOVEMENT TO COVER	ARRIVES AFTER YOU COMMIT
At its pitch	405 ms	215 mm	51 mm
2.5 m down the pitch	495 ms	287 mm	79 mm
At the crease	546 ms	334 mm	92 mm
Back, at the stumps	592 ms	379 mm	102 mm

- Standing deep buys about **97 ms** more watching time.
- It also doubles the late movement — 51 mm becomes 102 mm. A cricket ball is 71 mm across, so that is the difference between the middle of the bat and an edge.

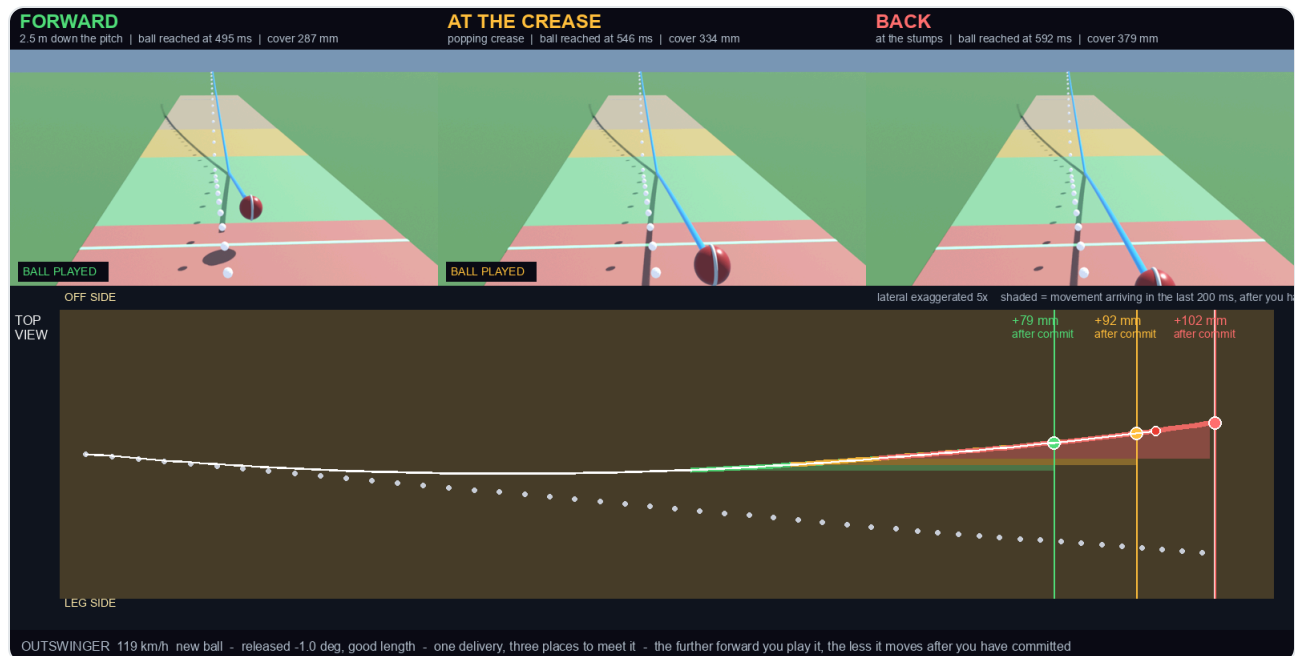


Figure 3. Each panel freezes when the ball reaches that point. The shaded wedges are the movement still to come.

THE TAKEAWAY

Judge the line. Meet it early.

I went looking for a way to predict the swing and found there was nothing to predict — it is very nearly the same every ball. What varies is where the bowler aimed, and how much of the movement is still ahead of me when I commit.

- Read the release and the pitch, then allow about a ball and a half to the off side.
- Meet it as close to its pitch as you can — guard outside the crease, a full stride, or both. That halves the late movement.
- Do not play across it. Bat and ball are already diverging.
- Stop treating "it swung late" as the explanation. It swung all the way.

[More field notes at systemsnotcode.com](https://systemsnotcode.com) →



Method & limits

Every number here comes from one configuration: a new ball at 119 km/h, seam angled 20° , released 2.2 m up. Read the limits before you trust the conclusions.

- **Movement off the seam at the bounce is not modelled.** The ball is a smooth sphere, so the pitch gives bounce and friction but cannot catch a raised seam. That effect *is* length-dependent, and it is the one thing that would undercut Chapter 1.
- **Aerodynamic swing only** — drag, a seam-driven side force, and a Magnus force from backspin.
- **Surface state is a single number**, set here to a new ball. Reverse swing exists in the model but is switched off for this study.
- **The 200 ms commit window is my assumption**, not a measurement. It sets the "arrives after you commit" column; change it and those numbers scale.
- **The model only says *where* the ball is met, not how you get there.** Taking guard outside the crease and striding to the pitch reach the same point. It says nothing about what standing out costs you — being stumped, or handing the bowler a different effective length.
- **The side-force coefficients are semi-empirical** — chosen to reproduce observed magnitudes, not fitted to wind-tunnel data.

HOW IT WAS BUILT

1. Ball flight and bounce as a Modelica model — multibody dynamics with a contact model for the pitch, solved in Wolfram System Modeler.
2. Aerodynamics as a custom component: drag, seam side force, Magnus lift, fed by the ball's own velocity and spin.
3. Trajectories exported and rendered in Blender, driven headlessly from the simulated data. Nothing in the animations is hand-animated.
4. The dotted line in every figure is the same delivery with the side force set to zero — the honest control.